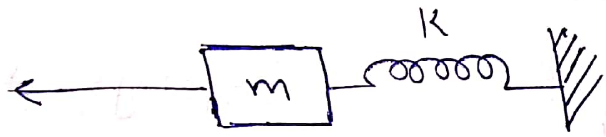
DOM

MODULE - 5

Mass Spring System

Natural frequency of oscillation using energy method



Sum of energy oscillated with system = Constant

$\therefore$ K.E + P.E = Constant

$\frac{1}{2} m v^2 + \frac{1}{2} K x^2 =$ constant

Substitute $v = \dot{x}$

$$\frac{1}{2} m \dot{x}^2 + \frac{1}{2} K x^2 = C$$

Derivating on both sides,

$$\frac{d}{dt}\left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} K x^2\right) = \frac{d}{dt}(c)$$

$$\frac{1}{2}\left(2 m \dot{x} . \ddot{x} + 2 K x \dot{x}\right) = 0$$

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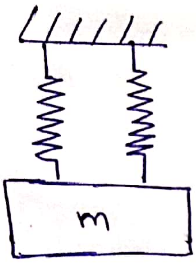
$$m\ddot{x} + Kx = 0$$

$$-\omega^2 m + K = 0$$

$$\boxed{\omega = \sqrt{K/m}}\ \text{rad/sec} \quad \text{and} \quad f = \frac{1}{2\pi}\sqrt{K/m}\ \text{Hz}$$

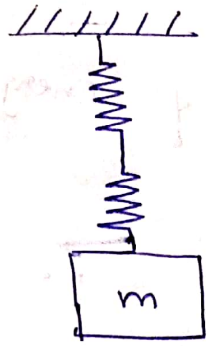
Equivalent stiffness of spring combinations

Springs in parallel



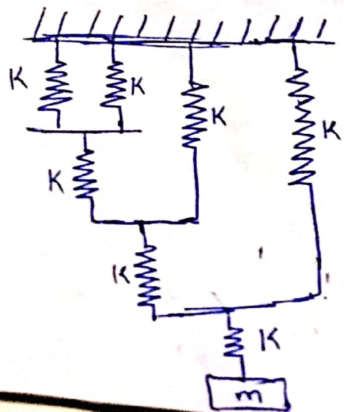
$$K_e = K_1 + K_2$$

Springs in series

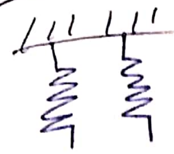


$$K_e = \frac{K_1 \cdot K_2}{K_1 + K_2}$$

Q) Find K_e & Material frequency (Take $K = 2\times10^5$ & $m = 20$



(a) Parallel

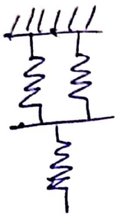


$K_1 = 2\times10^5$

$K_2 = 2\times10^5$

$K_e = K_1 + K_2 \Rightarrow 2\times10^5 + 2\times10^5 = \underline{400000}$

Series

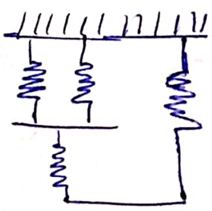


$K_1 = 400000$

$K_2 = 2\times10^5$

$$K_e = \frac{K_1 \times K_2}{K_1 + K_2} = \underline{133333.3}$$

Parallel



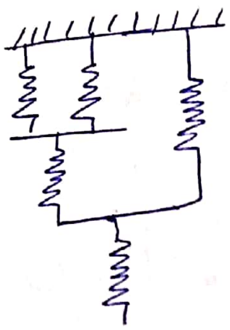
$K_1 = 133333.3$

$K_2 = 2\times10^5$

$K_e = 133333.3 + 2\times10^5$

$= 333333.3$

Series

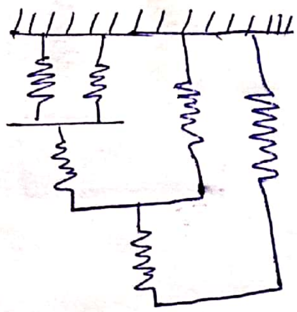


$K_1 = 333333.3$

$K_2 = 2 \times 10^5$

$$K_e = \frac{K_1 K_2}{K_1 + K_2} = 124999.9$$

Parallel



~~Ke~~ $K_1 = 124999.9$

$K_2 = 2 \times 10^5$

$$K_e = K_1 + K_2 = 124999.9 + 2 \times 10^5$$

$$K_e = 324999.9$$

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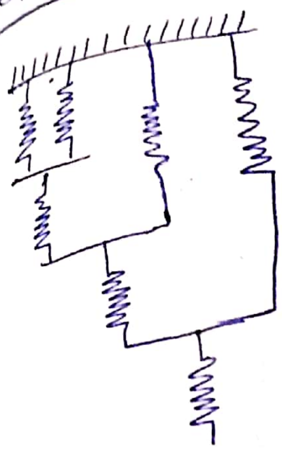
series



$K_1 = 324999.9$

$K_2 = 2 \times 10^5$

$$K_e = \frac{K_1 \cdot K_2}{K_1 + K_2}$$

$$K_e = 123809.5 \text{ N/m}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{123809.5}{20}}$$

$$f = 12.52 \text{ Hz}$$

$$\omega = \sqrt{\frac{K}{m}}$$

$$= \sqrt{\frac{123809.5}{20}}$$

$$= 78.67 \text{ rad/sec}$$

NOTE :

(i) If roots are different D_1 , D_2

$$x = A_1 . e^{D_1 t} + A_2 . e^{D_2 t}$$

(ii) If roots are same D_1 , D_1

$$x = A_1 e^{D_1 t} + t . A_2 e^{D_1 t}$$

(iii) If roots are complex conjugate $\alpha \pm i\beta$

$$x = e^{\alpha t} (A_1 \cos \beta t + A_2 \sin \beta t)$$

Critical Damping Constant & Damping ratio

When damping coefficient c is such that

$\left(\frac{c}{2m}\right)^2 - \frac{K}{m} = 0$ is known as critical damping const

Natural frequency

$$\omega = \sqrt{K/m} \quad -①$$

On solving,

$$C_c = 2m\sqrt{\frac{K}{m}}$$

$$\boxed{C_c = 2m\omega} \quad — ②$$

$C_c \to$ critical damping coefficient

The ratio c to C_c is damping ratio / damping factor

$\xi \to$ Damping factor

$$\boxed{\xi = \frac{c}{C_c} \Rightarrow \frac{c}{2m\omega}} \longrightarrow ③$$

(Substitute in eq ①)

$$x = A_1 e^{\left[-\xi + \sqrt{\xi^2 - 1}\right]\omega t} + A_2 e^{\left[-\xi - \sqrt{\xi^2 - 1}\right]\omega t}$$

Depending on ξ, the damped system are of 2 types

(i) Over damped $(\xi > 1)$

$$x = A_1 e^{\left[-\xi + \sqrt{\xi^2 - 1}\right]\omega t} + A_2 e^{\left[-\xi - \sqrt{\xi^2 - 1}\right]\omega t}$$

The motion is aperiodic

(ii) Critically damped system $(\xi = 1)$

$\xi = 1$, then $u_1 = u_2 = \frac{-c}{2m} = -\omega.$

$$x = A_1 e^{-\omega t} + t A_2 . e^{-\omega t}$$

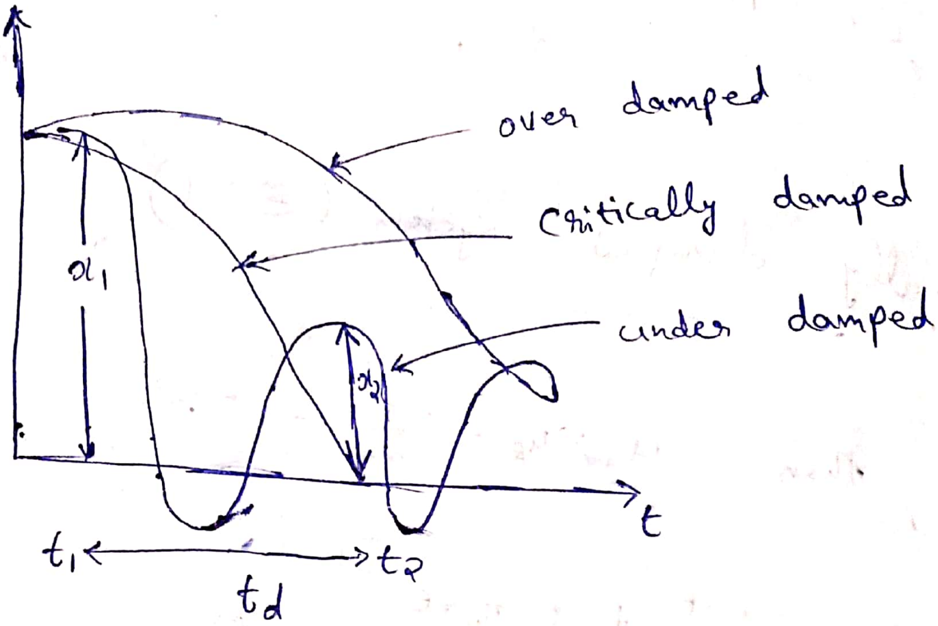
(iii) Under damped system $(\xi < 1)$

$$u_1 = \left(-\xi + i\sqrt{1 - \xi^2}\right)\omega$$

$$u_2 = \left(-\xi - i\sqrt{1 - \xi^2}\right)\omega$$

$$x = e^{-\xi\omega t}\left[A_1 . \cos\sqrt{1 - \xi^2} . \omega t + A_2 \sin\sqrt{1 - \xi^2} . \omega t\right]$$

Graph



Logarithmic decremant

Logarithmic decremant is defined as the natural logarithm of the ratio of any two successive amplitude on the same side of the mean time

$$\delta = \ln \frac{x_1}{x_2}$$

Let $t_2 = t_1 + t_d$,

$$t_d = \frac{2\pi}{\omega d}$$

$t_d \rightarrow$ Period of damp vibration

$$\frac{x_1}{x_2} = \frac{e^{-\varepsilon\omega t_1}}{e^{-\varepsilon\omega t_2}}$$

so, $\frac{x_1}{x_2} = e^{-\varepsilon\omega(t_1 - t_2)}$

$$= e^{-\xi\omega(t_1 - t_1 - t_d)} \qquad | \; t_2 = t_1 + t_d$$

$$= e^{\xi\omega t_d}$$

$$\boxed{\frac{x_1}{x_2} = e^{\xi\omega \frac{2\pi}{\omega_d}}}$$

$$\boxed{\omega_d = \left(\sqrt{1-\xi^2}\right)\omega} \quad \text{—— (4)}$$

Damped frequency

$$\frac{x_1}{x_2} = e^{\frac{\xi \, 2\pi}{\sqrt{1-\xi^2}}}$$

$$\boxed{\delta = \ln\frac{x_1}{x_2} = \frac{2\pi\xi}{\sqrt{1-\xi^2}}} \quad \text{—— (5)}$$

Logarithmic decrement

For 'n' oscillations,

$$\boxed{\delta = \frac{1}{n}\ln\frac{x_n}{x_{n+1}}} \quad \text{—— (6)}$$

Q) A vibrating system in a vehicle is designed as

$K = 100\ N/m$, $C = 2\ Ns/m$, $m = 1\ kg$

a) Calculate the decrease of amplitude from starting value of ter 3 complete oscillations

b) Frequency of oscillations (damped frequency)

Ans)

a) ~~8~~ ~~3~~ From eq ⑥, $n = 3$

$$\delta = \frac{1}{n} \ln \frac{x_1}{x_2}$$

$$\delta = \frac{1}{3} \ln \frac{x_1}{x_2}$$

$$\delta = \frac{2\pi \varepsilon}{\sqrt{1-\varepsilon^2}}$$

$$\varepsilon = \frac{c}{2m\omega}$$

$$\omega = \sqrt{K/m} = \sqrt{\frac{100}{1}} = 10\ \text{ra}$$

So,

$$\varepsilon = \frac{c}{2m\omega}$$

$$= \frac{2}{2 \times 1 \times 10} = \frac{1}{10} = 0.1$$

$$\delta = \frac{2\pi \varepsilon}{\sqrt{1-\varepsilon^2}} = \frac{2\pi \times 0.1}{\sqrt{1-(0.1)^2}} = 0.6315$$

from the eq,

$$\delta = \frac{1}{3} \ln \frac{x_1}{x_2}$$

$$0.6315 = \frac{1}{3} \ln \frac{x_1}{x_2}$$

$$0.6315 \times 3 = \ln \frac{x_1}{x_2}$$

$$\ln \frac{x_1}{x_2} = 1.89 \implies \frac{x_1}{x_2} = e^{1.89}$$

$$\frac{x_1}{x_2} = 6.62$$

b) $$\omega_d = \left(\sqrt{1-\xi^2}\right)\omega \longrightarrow \text{Eq. } (4)$$

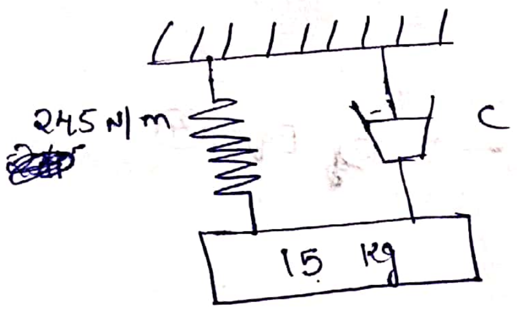
$$= \left(\sqrt{1-(0.1)^2}\right) \times 10$$

$$= 9.95 \text{ rad/sec}$$

Q) For the system shown the characteristics of the dashpot is such that when a constant force of 49 N is applied to the piston if ~~co~~ constant at velocity is found to be 0.12 m/s

(a) Determine the value of c

(b) would you expect the complete system to be periodic or aperiodic



ans)

a) $F = C \times \dot{x}$

$F = 49\,N$
$\dot{x} = 0.12\ m/s$

$$C = \frac{49}{0.12} = 408.33\ Ns/m$$

b) $$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{245}{15}}$$

$$\omega = 4.041\ rad/sec$$

$C_c = 2m\omega$

$= 2 \times 15 \times 4.041$

$= 121.23$

$\varepsilon = \frac{C}{C_c} = \frac{408.33}{121.23}$

$\varepsilon = 3.36 > 1$ over damped so aperiodic

Q) A damper offer resistance of 0.05 N at constant velocity 0.04 m/s. Determine the damping frequency of the system when m = 0.1 kg and K = 9 N/m

ans) $F = C \times \dot{x}$

$0.05 = C \times 0.04$

$C = 1.25$ N s/m

$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{9}{0.1}}$

$= 9.5$ rad/sec

$\omega_d = \left(\sqrt{1-\varepsilon^2}\right)\omega$

$\Rightarrow \varepsilon = \frac{C}{2m\omega} = \frac{1.25}{2 \times 0.1 \times 9.5} \Rightarrow 0.657$

$\omega_d = \left(\sqrt{1-(0.657)^2}\right) \times 9.5 \Rightarrow 7.16$ rad/sec

Q) A coil of spring stiffness 4 N/mm supports vertically a mass of 20 Kg at the free end. The motion is resisted by the oil dashpot. It is found that the amplitude at the beginning of the fourth cycle is 0.8 times the amplitude of previous vibration. Determine the damping force / unit velocity. Also find the ratio of frequency of damped and undamped vibration.

(An) $K = 4 \ N/mm$

$\Rightarrow \frac{4}{10^{-3}} \ N/m \Rightarrow 4 \times 10^3 \ N/m$

$m = 20 \ Kg$

$\omega = \sqrt{\frac{K}{m}} \Rightarrow \sqrt{\frac{4 \times 10^3}{20}}$

$\omega = 14.14 \ rad/sec$ (undamped)

$\left. \begin{array}{l} \text{Previous} \\ \text{adjacent} \\ \text{Consecutive} \end{array} \right\} n = 1$

Damping force / unit velocity, $c = ?$

0.

a)

$$\xi = \frac{c}{2m\omega}$$

$$C = \xi \times 2m\omega \quad \text{—①}$$

$$(x_2 = 0.8x_1)$$

$$\delta = \frac{2\pi\xi}{\sqrt{1-\xi^2}} \quad \text{—②}, \qquad \delta = \frac{1}{n} \ln \frac{x_1}{0.8x_1}$$

$$\Rightarrow \frac{1}{1} \cdot \ln \frac{1}{0.8}$$

$$= 0.2231$$

Sub in eq②

$$\delta = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$$

$$0.2231 = \frac{2 \times \pi \times \xi}{\sqrt{1-\xi^2}}$$

$$\xi = 0.035$$

a) Damping force / unit velocity, C

$$\cancel{C = \xi \times 2m\omega} \quad C = \xi \times 2m\omega$$

$$= 0.035 \times 2 \times 20 \times 14.14$$

$$C = 19.796 \ Ns/m$$

b) Ratio of damped & undamped

$$\omega_d = \left(\sqrt{1-\varepsilon^2}\right) \times \omega$$

$$= \sqrt{1-(0.035)^2} \times 14.14$$

$$= 14.13 \text{ rad/sec}$$

$$\omega_n = 14.14 \text{ rad/sec}$$

$$\frac{\omega_d}{\omega_n} = \frac{14.13}{14.14}$$

$$= 0.999$$

Q) A machine of mass 75 Kg is mounted on springs and is fitted with a dashpot to damp out vibrations. There are three springs each of stiffness 10 N/mm and it is found that the amplitude of vibration diminishes from 38.4 mm to 6.4 mm in two complete oscillations. Assuming that the damping force varies as the velocity, determine

(i) The resistance of the dashpot at unit velocity

(ii) The ratio of the frequency of the damped vibration to the frequency of the undamped vibration

(iii) The periodic time of the damped vibration

an) The no. of springs = 3

K in each spring = 10 N/mm $\Rightarrow 10\times10^3$ N/m

Total $K = 3\times 10\times10^3$

$= 30000$ N/m

(i) $m = 75$ Kg

$x_1 = 38.4$ mm

$x_2 = 6.4$ mm

$\omega = \sqrt{K/m}$

$= \sqrt{\frac{30000}{75}}$

$= 20$ rad/sec

$c = 2m\omega_n \varepsilon$ — ①

$\delta = \frac{1}{n} \ln \frac{x_1}{x_2}$ — ②

$n = 2$

$\delta = \frac{1}{2} \ln \frac{38.4}{6.4}$

$\delta = 0.89$

$\delta = \frac{2\pi \varepsilon}{\sqrt{1-\varepsilon^2}}$

$0.89 = \frac{2\pi \varepsilon}{\sqrt{1-\varepsilon^2}}$

$\varepsilon = 0.14$

sub in eq ①

$c = 2m\omega_n \varepsilon$

$= 2 \times 75 \times 20 \times 0.14$

$c = 420$ Ns/m

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(ii) $\omega_d = (\sqrt{1-\varepsilon^2}) \times \omega$

$= \sqrt{1 - 0.14^2} \times 20$

$= \underline{19.8}$ rad/sec

$\omega_n = 20$ rad/sec

$\frac{\omega d}{\omega_n} = \frac{19.8}{20}$

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$\frac{\omega_d}{\omega_n} = 0.99$

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a(iii) The periodic time, $t_d = \frac{2\pi}{\omega_d}$

$t_d = \frac{2\pi}{19.8}$

$t_d = 0.31$ sec

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